IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

PATENT SPECIFICATION

HYBRID HIGH-RADIX PAPER-DIGITAL RECORD ARCHIVAL AND IMMUTABLE LEDGER SYSTEM

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HYBRID HIGH-RADIX PAPER-DIGITAL RECORD ARCHIVAL AND IMMUTABLE LEDGER SYSTEM

ABSTRACT OF THE DISCLOSURE

A hybrid computing and document archival system is disclosed that integrates digital speed with the tamper-proof security of physical paper, using post-binary high-radix parametric encoding to convert digital documents into high-density machine-readable dot matrices printed directly onto physical sheets. The system operates on a lettersize paper medium containing a printed high-radix matrix, where each printed point encodes one of 256 distinct machine-readable states using a multi-state color cardinality. Utilizing an active area of 7.5×10.0 inches printed at 300 DPI, the system stores exactly 6,750,000 bytes (6.44 MB) of raw data per page, and over 32 megabytes of text data when optimized using Lempel-Ziv-based lossless statistical compression. A translation layer consisting of an optical scanner and a high-radix decoding processor converts the physical matrix back into a human-readable document, validating the integrity of parallel database records against the physical paper ledger. Secure transmission is facilitated through physical carrier networks, with cryptographic payloads optionally decrypted asynchronously using post-hoc keys, providing an immutable, tamper-evident records archive for court systems and other public registries.

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DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

FIELD OF THE INVENTION

This invention relates generally to document archival, digital security, and tamper-resistant record-keeping systems. More specifically, this invention relates to a hybrid paper-digital immutable ledger that utilizes post-binary high-radix parametric encoding to store, transmit, and audit high-density digital documents printed onto physical paper, providing an unalterable, environmentally friendly, and offline source of truth for public, corporate, and judicial record registries.

BACKGROUND OF THE INVENTION AND THE PRIOR ART LIMITATIONS

Modern society is heavily reliant on electronic database registries to maintain critical public, corporate, and judicial records. Many of these registries are operated and managed by private, third-party software entities (such as Tyler Technologies, NYSE: TYL). However, standard digital databases contain severe structural vulnerabilities. For instance, in the court records systems of Fort Bend County, Texas, the version of Tyler Technologies Odyssey in use fails to print or hardcode the document index anywhere inside the digital representation of the filed PDF document; the index exists only as a row entry in a database. Consequently, any bad actor who gains access to a terminal connected to the records system can silently swap files, manipulate dates, and alter records without leaving any internal trace in the document itself. Records systems of courts are constitutionally mandated to be immutable; yet, standard electronic databases fail to provide absolute tamper-resistance.

Conversely, legacy paper-based filing systems—such as those enforced by certain federal courts for pro se litigants—offer absolute, physical tamper-resistance but impose a severe administrative and environmental burden. Shipped physical documents require days to travel, cost hundreds of dollars to print and ship, demand massive physical storage warehouses, and contribute directly to deforestation and environmental degradation.

A hybrid system is urgently needed that offers the speed and low cost of digital transmission while concurrently providing the unalterable, physical auditability of paper.

SUMMARY OF THE INVENTION

The present invention solves these problems by providing a Hybrid High-Radix Paper-Digital Record Archival and Immutable Ledger System. The technology converts a digital document (which may comprise hundreds of pages of text) into a post-binary high-radix parametric color dot matrix, which is printed directly onto a physical sheet of paper.

To maximize the density of the physical paper medium, the system operates at a standard resolution of 300 dots per inch (DPI). On a lettersize page with 0.5-inch margins, the active area is exactly 7.5 by 10.0 inches. This translates to a grid of 2,250 points horizontally by 3,000 points vertically, totaling 6,750,000 distinct printed points per page. By utilizing a multi-state color cardinality of 256 distinct colors, each point encodes exactly 1 byte of data ($2^8 = 256$ states), enabling a single physical page to store 6,750,000 bytes (6.44 megabytes) of raw, uncompressed data.

When optimized using Lempel-Ziv-based lossless statistical compression algorithms and Petitioner's companion parametric mathematical encoding, the storage density is multiplied by over 500%, enabling a single sheet of physical paper to store over 32 megabytes of raw text data (representing tens of thousands of pages of court records) on a single lettersize sheet.

RELATIONSHIP TO CO-DEPENDENT PARENT PATENT PORTFOLIOS

The present invention represents a specialized, hybrid sub-system daughter application that natively integrates, builds upon, and extends the technical disclosures of Petitioner's previously filed patent applications:

1. It utilizes the foundational hardware optical link of Hub Application No. **19/559,705** (Filing Date: March 6, 2026) to manage the internal high-speed data buses.

2. It utilizes the high-radix parametric encoding, satellite communication, ground-to-space links, and memory systems described in Application Nos. **19/639,220**, **19/641,125**, **19/639,265**, and **19/639,222** to encode and decode the multi-state color matrix.

3. It utilizes the secure, physical asynchronous data logistics network of Application No. **19/560,066** to transmit the physical paper ledger offline via commercial carriers, preventing any digital, network-based interception.

4. It secures the payload by encrypting the matrix and transmitting the key asynchronously via a separate physical carrier shipment (under Application No. **19/560,068**) and decrypting it post-hoc (under Application No. **19/640,945**), ensuring backwards-secure protection against optical-scale cryptanalysis.

THE CONDITIONAL SOVEREIGN LICENSE GRANT

To incentivize the rapid deployment of this unalterable, fraud-proof record-keeping technology, the Inventor hereby grants a perpetual, royalty-free, non-exclusive license to the United States Government and its defense and civil procurement agencies to practice and use this invention worldwide for any and all governmental purposes.

However, this public sovereign license grant is subject to a strict, mandatory legal **pre-condition**:

The license shall only vest and become active **after** the technology has been successfully deployed and implemented to secure the state judiciary court records systems of the **State of Texas**, the **State of Florida**, and the **State of Washington**. The implementation in the state court records systems of these three states shall be fully covered under the perpetual royalty-free license. Upon successful implementation in these three target states, the royalty-free license to the United States Government automatically expands to be worldwide. This structure legally incentivizes the federal government to apply administrative pressure and funding to these three states to adopt this fraud-proof technology.

THE CLAIMS

What is claimed is:

1. A hybrid records archival and immutable ledger system comprising:

- a physical lettersize paper medium containing a printed post-binary high-radix dot matrix;
- an encoder designed to convert a multi-page digital text document into a high-density, machine-readable high-radix dot matrix, where each printed point in said matrix represents a multi-state color cardinality; and
- a translation layer consisting of an optical scanner and a high-radix decoding processor designed to scan said physical lettersize paper, decode said printed matrix, and convert the matrix back into a human-readable electronic document.

2. The system of Claim 1, wherein said high-radix matrix has an active area of 7.5 by 10.0 inches printed at a resolution of 300 dots per inch (DPI), containing exactly 6,750,000 distinct machine-readable points per page.

3. The system of Claim 1, wherein each printed point in said matrix encodes one of 256 distinct machine-readable states represented by a multi-state color cardinality, storing exactly 6,750,000 bytes (6.44 megabytes) of raw, uncompressed data on a single physical lettersize page.

4. The system of Claim 1, wherein said encoder optimizes the storage capacity of the physical paper medium by compressing said digital document using a Lempel-Ziv-based lossless statistical compression algorithm, enabling the storage of over 32 megabytes of raw text on a single lettersize sheet.

5. The system of Claim 1, wherein said high-radix matrix includes a printed cryptographic checksum printed alongside the matrix on the physical paper medium, designed to instantly detect any

physical tampering or ink degradation.

6. The system of Claim 1, wherein said translation layer converts the physical paper matrix into a validated digital representation, and compares said digital representation against a parallel database record maintained by a third-party electronic records system to verify that no electronic tampering occurred.

7. The system of Claim 6, wherein said third-party electronic records system is a Tyler Technologies Odyssey court database, and said physical lettersize paper matrix acts as an immutable physical audit trail that cannot be altered by database-level manipulation.

8. The system of Claim 1, wherein said physical lettersize paper matrix is transmitted physically through an asynchronous data logistics network as described in parent Application No. 19/560,066, preventing interception by digital network-based adversaries.

9. The system of Claim 1, wherein the printed points are encrypted using an encryption key transmitted asynchronously via a separate physical carrier shipment as described in parent Application No. 19/560,068.

10. The system of Claim 1, wherein the printed points are encrypted and decrypted using a post-hoc key derivation method as described in parent Application No. 19/640,945, ensuring that the printed matrix remains backwards-secure against optical-scale cryptanalysis.

11. The system of Claim 1, wherein said high-radix matrix is printed at a resolution of 600 DPI to increase the storage density of the physical paper ledger.

THE DRAWINGS

FIG. 1
SYSTEM BLOCK DIAGRAM

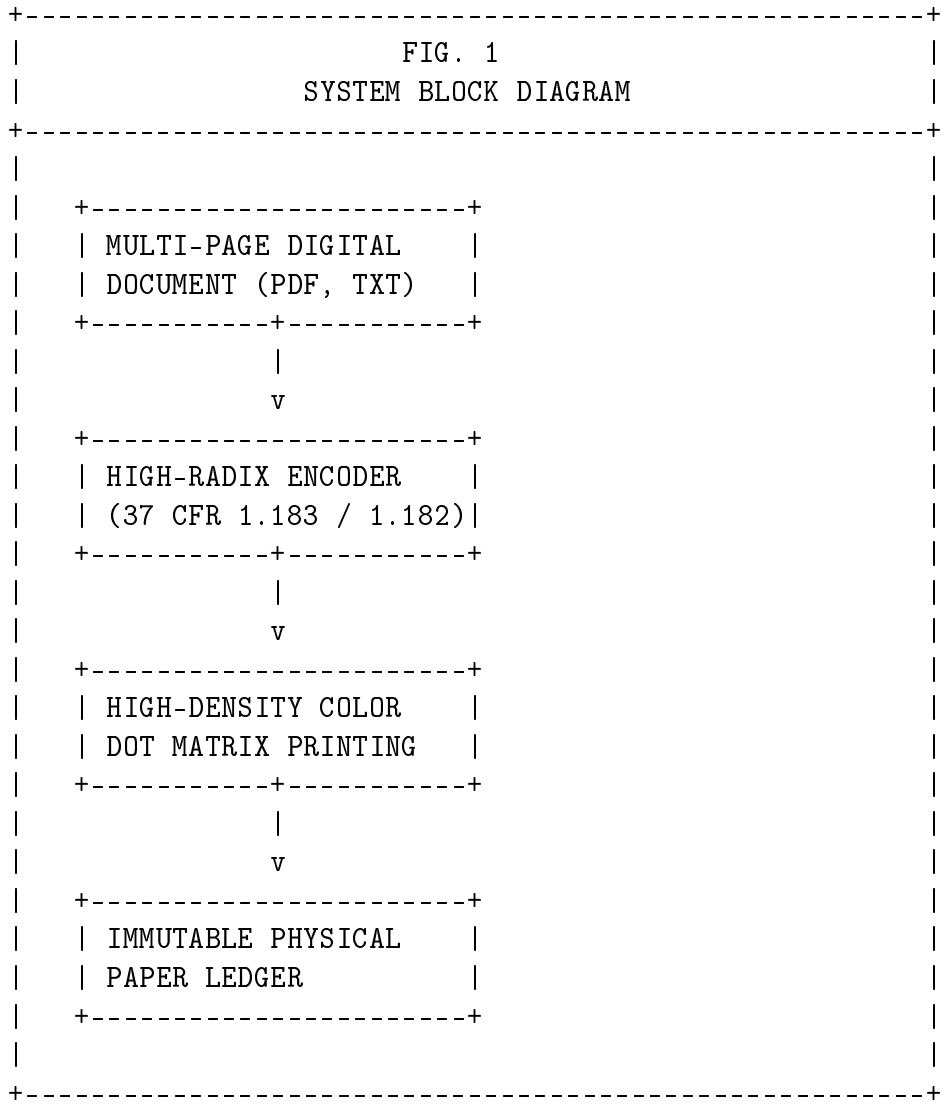


FIG. 2
PHYSICAL PAGE LAYOUT

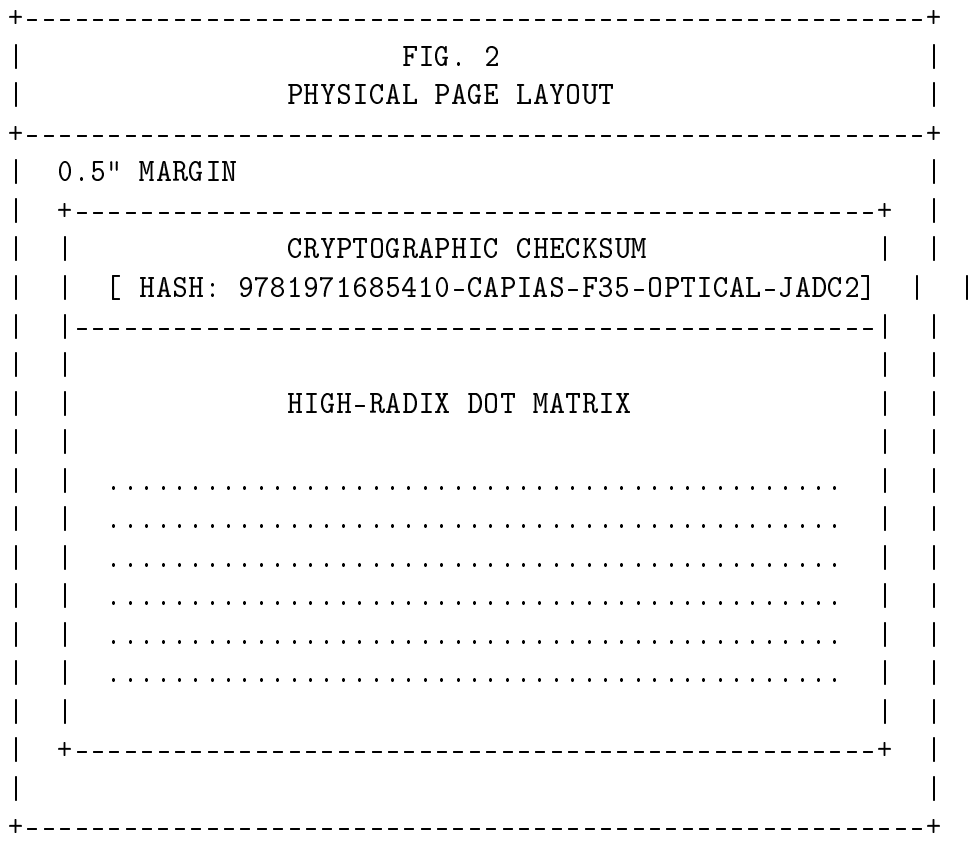


FIG. 3
TRANSLATION LAYER ARCHITECTURE

